

Aashish Rai

Providence, Rhode Island

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EDUCATION

- **Brown University** **Providence, RI**
Ph.D. in Computer Science
Advisor: Srinath Sridhar
Aug 2023 – May 2028
- **National Institute of Technology** **Surat, India**
Bachelor of Technology in Electronics and Communication
Aug 2017 – June 2021

RESEARCH EXPERIENCE

- **Brown University** **Providence, RI**
Graduate Research Assistant
Aug 2023 – Present
Advisor: Srinath Sridhar
 - Leveraging large reconstruction models and video models to build a feed-forward generative world model for 3D/4D scenes.
 - Built a novel 4D volumetric video representation (PackUV) that enables streaming via existing multimedia infrastructure. It packs 3D Gaussian attributes into a video of UV images for efficient streaming and storage (CVPR 2026).
 - Captured the largest multi-view video dataset (2 billion frames) with large motions, and disocclusions. It provides 360 coverage from 55-90 synchronized cameras with variable FPS (PackUV-2B).
 - Proposed EgoSonics, a conditional diffusion-based framework to generate semantically aligned and synchronized audio tracks for silent egocentric videos (WACV 2025).
- **Meta Reality Labs** **Burlingame, CA**
Researcher/Intern
May 2024 – Dec 2024
Host: Aayush Prakash
 - Proposed **UVGS**, a novel 2D representation to solve permutation invariance and the unstructured nature of 3D Gaussian Splatting primitives.
 - Enabled the direct application of existing 2D image generative models (VAEs, Diffusion) to 3D Gaussian Splatting, significantly reducing training overhead (CVPR 2025).
- **Carnegie Mellon University** **Pittsburgh, PA**
Research Assistant
Sept 2021 – May 2023
Advisor: Fernando De la Torre
 - **Generative 3D Faces:** Developed a framework to generate photorealistic 3D faces by leveraging 2D generative priors. Demonstrated applications in semantic manipulation and text-based 3D editing.
 - Achieved State-of-the-Art (SOTA) performance in 3D shape reconstruction and identity preservation (WACV 2024).
 - **Disentangled Representations:** Built a 3D face generative model to decouple identity from expression, enabling granular control over facial animation (WACV 2023).
- **McGill University** **Montreal, Canada (Remote)**
Research Intern
May 2020 – Mar 2021
Advisor: Jeremy Cooperstock
 - Improved Semantic Face Editing by manipulating the latent space of StyleGAN2.
 - Proposed an automated method for feature disentanglement in latent space via orthogonal projection (ICMLA 2021).
- **Norwegian Biometrics Laboratory, NTNU** **Norway (Remote)**
Undergraduate Researcher
Dec 2019 – May 2020
Advisors: Kishor Upla, Christoph Busch
 - Designed an efficient face super-resolution model using a progressive residual CNN network (IWBF 2020).
 - Outperformed benchmark datasets CelebA (PSNR: 26.55) and LFW (PSNR: 26.26).

PUBLICATIONS

- **A. Rai**, A. Xing, A. Agarwal, X. Cong, Z. Li, T. Lu, A. Prakash, S. Sridhar. "*PackUV: Packed Gaussian UV Maps for 4D Volumetric Video*", **CVPR 2026**. [Project Page]
- **A. Rai**, D. Wang, M. Jain, N. Sarafianos, A. Chen, S. Sridhar, A. Prakash. "*UVGS: Reimagining Unstructured 3D Gaussian Splatting using UV Mapping*". **CVPR 2025**. [Project Page]
- **A. Rai**, S. Sridhar. "*EgoSonics: Generating Synchronized Audio for Silent Egocentric Videos*". IEEE/CVF Winter Conference on Applications of Computer Vision, **WACV 2025**. [Project Page]
- **A. Rai**, H. Gupta, A. Pandey, F. V. Carrasco, S. J. Takagi, A. Aubel, D. Kim, A. Prakash, F. de la Torre. "*Towards Realistic Generative 3D Face Models*". **WACV 2024**. [Project Page]
- F. Teherkhani, **A. Rai**, S. Srivastava, Q. Gao, X. Chen, F. de la Torre, S. Song, A. Prakash, D. Kim. "*Controllable 3D Generative Adversarial Face Model via Disentangling Shape and Appearance*". **WACV 2023**. [Project]
- **A. Rai**, C. Ducher, J. Cooperstock. "*Improved Attribute Manipulation in the Latent Space of StyleGAN for Semantic Face Editing*". **ICMLA 2021**. [Paper]
- **A. Rai**, V. Chudasama, K. Upla, K. Raja, R. Ramachandra, C. Busch. "*ComSupResNet: A Compact Super-Resolution Network for Low-Resolution Face Images*". **IWBF 2020**. (Extended version in IEEE Transactions on Biometrics). [Paper]

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C++, CUDA, Parallel Programming, PyTorch, NumPy
- **3D/4D Vision:** Gaussian Splatting (3DGS, 4DGS), NeRF, Neural Rendering, Volumetric Reconstruction, Multiview Geometry, Camera Estimation, Differentiable Rendering, Optical Flow, COLMAP
- **Generative AI:** Diffusion Models (DDPM, DDIM, 3D Diffusion), World Models, Large Language Models (LLMs), Multimodal Learning, GANs, Face Generation, Video Generation
- **Core Competencies:** 3D/4D Computer Vision, Stereo & Volumetric Reconstruction, Free-viewpoint Video, Deep Learning, Large Dataset Curation